

Tianyu Chen

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🔗 Google Scholar

🔗 TianyuCodings

EDUCATION

University of Texas at Austin	Ph.D. in Statistics (GPA: 4.0), Austin, TX	Aug. 2023 – May 2028 (expected)
University of Chicago	M.S. in Statistics (GPA: 3.97), Chicago, IL	Oct. 2021 – Jun. 2023
Fudan University	B.S. in Statistics (GPA: 3.7), Shanghai, China	Sep. 2017 – Jul. 2021

INDUSTRY EXPERIENCE

Google DeepMind | Gemini

Student Researcher

Mar. 2026 – Present

- Post-training the Gemini Embedding model for **agentic search**: turning it into an **autoregressive** retriever that iteratively issues queries for **multi-hop visual retrieval** over million-scale corpora far too large to fit into the model's context window.
- Designing the reinforcement learning (RL) algorithm that trains this retriever end-to-end from multi-hop retrieval outcomes, improving instruction following on compositional image retrieval tasks.

Microsoft Research | MAI Superintelligence

Research Scientist Intern (Mentors: Zhendong Wang, Lijuan Wang)

Feb. 2025 – Mar. 2026

- Developed an object-centric vision-language model (VLM) agent that automates fine-grained, multi-turn image-editing evaluation, reducing reliance on human annotators for benchmark construction; the work received media coverage in Chinese tech outlets (project website).
- Led large language model (LLM) post-training research on mathematical reasoning, LLM-as-a-judge alignment, and on-policy distillation, including training-recipe development for **autoregressive** and **diffusion-based** language models.

MORE INFORMATION

LinkedIn: <https://www.linkedin.com/in/tianyu-chen-1a056a160/>

Website: <https://tianyucodings.github.io/>

Research Interests: LLM post-training (RL and on-policy-distillation), image editing, and model distillation

GENERATIVE MODEL & REINFORCEMENT LEARNING

- **[Preprint] Tianyu Chen**, Jiaxing Wu. “Test-Time Optimization of Query Embeddings with Ranking Aware Reward Maximization.” Preprint (with Google DeepMind). [\[arXiv\]](#). 2026.
- **[Preprint] Jiahui Huang**, Yasi Zhang, **Tianyu Chen**, Shu Wang, Jianwen Xie, Oscar Leong, Mingyuan Zhou, Nanzhu Wang, Ying Nian Wu. “MT-EditFlow: Reinforcement Learning for Multi-Turn Image Editing with Flow Matching.” Preprint [\[arXiv\]](#). 2026.
- **[ICML 2026] Tianyu Chen***, Yasi Zhang*, Mingyuan Zhou, Oscar Leong, Ying Nian Wu, Michal Lukasik[†]. “REAL: Regression-Aware Reinforcement Learning for LLM-as-a-Judge.” Accepted to: *ICML 2026* (with Google DeepMind). [\[arXiv\]](#). 2026.
- **[NeurIPS 2024] Tianyu Chen**, Zhendong Wang, Mingyuan Zhou. “Diffusion Policies creating a Trust Region for Offline Reinforcement Learning.” Published in: *NeurIPS 2024*. [\[arXiv\]](#). 2024.
- **[ICLR 2026] Tianyu Chen***, Yasi Zhang*, Zhi Zhang, Peiyu Yu, Shu Wang, Zhendong Wang, Kevin Lin, Xiaofei Wang, Zhengyuan Yang, Linjie Li, Chung-Ching Lin, Jianwen Xie, Oscar Leong, Lijuan Wang, Ying Nian Wu, Mingyuan Zhou. “EdiVal-Agent: An Object-Centric Framework for Automated, Scalable, Fine-Grained Evaluation of Multi-Turn Editing.” Accepted to: *ICLR 2026*. [\[OpenReview\]](#). 2026.
- **[ICLR 2026] Tianyu Chen***, Yasi Zhang*, Zhendong Wang, Ying Nian Wu, Mingyuan Zhou, Oscar Leong. “Score Distillation Beyond Acceleration: Generative Modeling from Corrupted Data.” Accepted to: *ICLR 2026*. [\[OpenReview\]](#). 2026.
- **[Preprint] Tianyu Chen***, Xinran Song*, Mingyuan Zhou. “A Generative Framework for Causal Estimation via Importance-Weighted Diffusion Distillation.” Preprint [\[arXiv\]](#). 2025.
- **[NeurIPS 2025] Yifan Sun**, Jingyan Shen, Yibin Wang, **Tianyu Chen**, Zhendong Wang, Mingyuan Zhou, Huan Zhang. “Improving Data Efficiency for LLM Reinforcement Fine-tuning Through Difficulty-targeted Online Data Selection and Rollout Replay.” Accepted to: *NeurIPS 2025*. [\[arXiv\]](#). 2025.
- **[PNAS] Tianyu Chen***, Jin-Hong Du*, Ming Gao, Jingshu Wang. “Joint trajectory inference for single-cell genomics using deep learning with a mixture prior.” Published in: *PNAS*. [\[PNAS\]](#). 2024.

STATISTICS

- **[NeurIPS 2025 Spotlight] Tianyu Chen**, Vansh Bansal, James G. Scott. “CoLT: The conditional localization test for assessing the accuracy of neural posterior estimates.” **Spotlight at:** *NeurIPS 2025*. [\[arXiv\]](#). 2025.
- **[ICML 2026] Tianyu Chen***, Vansh Bansal*, James G. Scott. “Conformal C2ST: Turning Weak Classifiers into Strong Two-Sample Tests.” Accepted to: *ICML 2026*. [\[arXiv\]](#). 2025.
- **[AISTATS 2025] Tianyu Chen**, James G. Scott. “Conditional diffusions for neural posterior estimation.” [\[arXiv\]](#). 2025.
- **[NeurIPS 2024] Tianyu Chen**, Kevin Bello, Francesco Locatello, Bryon Aragam, Pradeep Ravikumar. “Identifying General Mechanism Shifts in Linear Causal Representations.” Published in: *NeurIPS 2024*. [\[arXiv\]](#). 2024.
- **[NeurIPS 2023] Tianyu Chen**, Kevin Bello, Bryon Aragam, Pradeep Ravikumar. “iSCAN: Identifying Causal Mechanism Shifts among Nonlinear Additive Noise Models.” Published in: *NeurIPS 2023*. [\[arXiv\]](#). 2023.